AutoCare AI: Agile Jira Workflow & Work Breakdown Structure

Formal Distribution across 3 Students (Talha, M. Abdullah, Abdullah) | 4 Sprints | 156 Story Points

Academic Scope:	PUCIT BS Data Science Final Year Project / Big Data Systems
Jira Key & Repo:	Project Key: **AUT** | Remote: github.com/marketbul/autocare-ai.git
Engineering Team:	Talha Ali Bhatti (016) | M. Abdullah (032) | Abdullah (035)
Velocity & Balance:	156 Story Points Total | Exactly 52 SP per member (100% Balanced)

1. Team Work Breakdown & RACI Responsibility Matrix

Epic / Module	Primary Lead	Talha (016)	M. Abdullah (032)	Abdullah (035)
EPIC-1: MinIO S3 & CAS Storage	Talha Ali Bhatti	**Accountable**	Consulted	Informed
EPIC-2: Consistent Hashing & Bloom Filter	Abdullah	Consulted	Informed	**Accountable**
EPIC-3: Stream Ingest & Parquet Lakehouse	Talha Ali Bhatti	**Accountable**	Consulted	Informed
EPIC-4: DuckDB Analytics & Fleet Anomaly	Muhammad Abdullah	Informed	**Accountable**	Consulted
EPIC-5: Causal Transformer LM & RAG Engine	Abdullah	Informed	Consulted	**Accountable**
EPIC-6: UI Web Portal, Benchmarks & Defense	Muhammad Abdullah	Consulted	**Accountable**	Consulted

2. 4-Week Sprint Schedule & Velocity Allocation

Sprint	Timeline & Focus	Talha (SP)	M. Abd (SP)	Abdullah (SP)	Total SP
Sprint 1	Days 1–7 (Storage & Consistent Hash)	13 SP	13 SP	13 SP	**39 SP**
Sprint 2	Days 8–14 (Bloom Filter & Stream Ingest)	13 SP	13 SP	13 SP	**39 SP**
Sprint 3	Days 15–21 (Transformer LM & DuckDB)	13 SP	13 SP	13 SP	**39 SP**
Sprint 4	Days 22–28 (RAG Integration & Web Portal)	13 SP	13 SP	13 SP	**39 SP**
Total	**4 Weeks (Platform Ready for Defense)**	**52 SP**	**52 SP**	**52 SP**	**156 SP**

3. Complete Jira Ticket Catalog & Commit Linkages (AUT-101 – AUT-132)

Key	Type	Summary & Commit Tag	SP	Assignee	Status
AUT-101	Story	**MinIO S3 Distributed Multi-Bucket Cluster Architecture** [AUT-101] feat(storage): provision distributed MinIO S3 cluster and bucket policies	**5**	Talha	**Done**
AUT-102	Story	**SHA-256 Content-Addressable Storage (CAS) Deduplication Engine** [AUT-102] feat(storage): implement SHA-256 content-addressable storage deduplication	**5**	Talha	**Done**
AUT-103	Story	**S3 Asynchronous Multipart Uploader with Exponential Backoff** [AUT-103] feat(storage): resilient asynchronous multipart uploader with exponential backoff	**3**	Talha	**Done**
AUT-104	Story	**Consistent Hash Ring with 160 Virtual Nodes per Physical Node** [AUT-104] feat(algorithms): consistent hash ring with 160 virtual nodes and Murmur3	**8**	Abdullah	**Done**
AUT-105	Story	**Consistent Hash Node Dynamic Rebalancing & Binary Ring Lookup** [AUT-105] feat(algorithms): bisect ring lookup and dynamic cluster rebalancing logic	**5**	Abdullah	**Done**
AUT-114	Story	**DuckDB High-Performance In-Memory OLAP Query Engine Integration** [AUT-114] feat(analytics): embed DuckDB OLAP engine for zero-copy parquet querying	**5**	Muhammad	**Done**
AUT-115	Story	**Automotive Dataset Ingestion & Entity Schema Standardization** [AUT-115] feat(data): clean, preprocess and standardize 111 automotive dataset files	**5**	Muhammad	**Done**
AUT-116	Story	**PySpark Distributed DataFrame Windowed Transformations Pipeline** [AUT-116] feat(analytics): PySpark distributed windowed transformations pipeline	**3**	Muhammad	**Done**
AUT-106	Story	**Counting Bloom Filter for O(1) Set Membership & Redundant I/O Gate** [AUT-106] feat(algorithms): counting bloom filter with k-hash double-hashing gate	**8**	Abdullah	**Done**
AUT-107	Story	**Bloom Filter Bit-Array Serialization & Redis State Sync** [AUT-107] feat(algorithms): bloom filter binary serialization and snapshot restore	**5**	Abdullah	**Done**
AUT-108	Story	**OBD-II CAN-Bus High-Velocity Telemetry Stream Generator** [AUT-108] feat(ingest): high-velocity OBD-II CAN-bus synthetic telemetry generator	**5**	Talha	**Done**
AUT-109	Story	**In-Memory Sliding-Window Batch Buffer & Ingestion Sink** [AUT-109] feat(ingest): sliding-window telemetry buffer and micro-batch flusher	**5**	Talha	**Done**
AUT-110	Story	**Snappy Columnar Parquet Lakehouse Partitioning (Make/Model/Year/Region)** [AUT-110] feat(lakehouse): snappy columnar parquet writer with hierarchical partitioning	**3**	Talha	**Done**
AUT-117	Story	**Fleet Diagnostic Trouble Code (DTC) Anomaly Detection Model** [AUT-117] feat(ml): unsupervised Isolation Forest anomaly detection for OBD-II telemetry	**8**	Muhammad	**Done**
AUT-118	Story	**Regional Component Degradation & Pakistani Climate Risk Multipliers** [AUT-118] feat(analytics): regional climate risk multipliers and component wear equations	**5**	Muhammad	**Done**
AUT-111	Story	**Automated Parquet Compaction Job & Small File Merging Engine** [AUT-111] feat(lakehouse): parquet small-file compaction and catalog vacuum engine	**5**	Talha	**Done**

AUT-112	Story	**Lakehouse Schema Evolution & Corrupt Record Dead-Letter Sink** [AUT-112] feat(lakehouse): schema evolution handler and dead-letter quarantine sink	**3**	Talha	**Done**
AUT-131	Task	**Distributed Lakehouse Cache Invalidation & Parquet Metadata Indexing** [AUT-131] feat(lakehouse): parquet footer metadata cache and partition indexer	**5**	Talha	**Done**
AUT-119	Task	**DuckDB Analytical Query Benchmark vs Traditional Relational SQL** [AUT-119] test(benchmark): comparative query latency benchmark of DuckDB vs SQLite	**5**	Muhammad	**Done**
AUT-120	Story	**Pakistani Fair-Price Market Estimator & Workshop Labor Models** [AUT-120] feat(pricing): localized PKR parts and mechanic labor fair-price estimator	**5**	Muhammad	**Done**
AUT-132	Story	**Workshop Diagnostic Estimate Verifier & Fraud Detection Heuristics** [AUT-132] feat(verifier): workshop estimate cross-checker and invoice fraud detector	**3**	Muhammad	**Done**
AUT-121	Story	**Custom Subword Byte-Pair Encoding (BPE) Tokenizer from Scratch** [AUT-121] feat(rag-scratch): subword BPE tokenizer with 4000 vocab and special tokens	**5**	Abdullah	**Done**
AUT-122	Story	**Dense L2-Normalized Vector Embedding Neural Network (D=128)** [AUT-122] feat(rag-scratch): dense L2-normalized vector embedding neural network (D=128)	**3**	Abdullah	**Done**
AUT-123	Story	**4-Layer Causal Transformer Decoder with Multi-Head Self-Attention** [AUT-123] feat(rag-scratch): 4-layer causal transformer decoder with multi-head attention	**5**	Abdullah	**Done**
AUT-124	Story	**Autoregressive LM Training Pipeline with AdamW & Perplexity Tracking** [AUT-124] feat(rag-scratch): autoregressive LM training pipeline with AdamW and Perplexity tracking	**5**	Abdullah	**Done**
AUT-125	Story	**Hybrid Vector Store (Cosine Similarity + BM25 Lexical Ranking)** [AUT-125] feat(rag-scratch): hybrid dense vector + BM25 lexical retrieval engine	**5**	Abdullah	**Done**
AUT-126	Story	**Multilingual Grounded RAG Generation & Anti-Hallucination Gate** [AUT-126] feat(rag-scratch): end-to-end grounded RAG generation with selective abstention	**3**	Abdullah	**Done**
AUT-113	Task	**Storage IOPS & Ingestion Latency Percentile Benchmark Profiler** [AUT-113] perf(storage): automated IOPS and ingestion latency benchmarking suite	**5**	Talha	**Done**
AUT-129	Task	**End-to-End System Integration Testing & Pytest Verification** [AUT-129] test(integration): full pytest test suite for distributed storage, hashing, and RAG	**8**	Talha	**Done**
AUT-127	Story	**Modern Automotive Diagnostic Web Portal Redesign (React + Tailwind)** [AUT-127] feat(frontend): modern automotive engineering web portal with diagnostic terminal	**5**	Muhammad	**Done**
AUT-128	Story	**FastAPI Asynchronous RAG Inference & Telemetry REST Endpoints** [AUT-128] feat(api): asynchronous RAG inference endpoints and Vite SPA static mount	**5**	Muhammad	**Done**
AUT-130	Story	**Final Academic Project Initiation Document (PID) & Defense Slides** [AUT-130] docs(academic): generate professional 4-week Big Data PID in PDF and Word	**3**	Muhammad	**Done**

4. Traceability & Automatic Jira Status Transitions

By enforcing the [AUT-xxx] commit message standard across all Git feature branches, the GitHub for Jira integration automatically advances stories from *In Progress* to *Code Review* and *Done*. Each member has completed exactly 52 Story Points, demonstrating balanced collaboration for academic evaluation.